

Saffron Livaccari

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PROFESSIONAL EXPERIENCE

Climate Data Scientist | ISS Stoxx | New York City, NY

June 2022 – Present

- Implemented company-level financial risk models for six natural hazards (river flood, coastal flood, wildfire, tropical cyclone, drought, and heat stress) with a team of three climate scientists
- Built scalable, cloud-based python workflows using Xarray, Dask, Zarr, and vectorized computations to process high resolution, multidimensional geospatial climate datasets over yearly time horizons
- Authored extensive technical documentation in Jira, Confluence, and inline-code comments

Project | Hazard Impact Database

- Technical lead on a climate risk pipeline translating physical risk data into financial risk metrics for SSP and NGFS scenarios using Google cloud VM and Google cloud storage
- Developed multiple classes for each hazard with memory-efficient processing and vectorized computation to calculate asset and country level damages and business interruptions
- Configured cloud bursting in google cloud to dynamically scale resources for intense, high-memory computations
- Quantified flood damage ratios using return-period interpolations, poisson probabilities, and trapezoid integrations

Project | Overhaul Legacy Pipeline

- Streamlined the physical risk corporate data pipeline to improve the maintainability and performance using xarray and dask
- Collaborated across teams to develop an automated and parallelized data pipeline using Docker containerization and Papermill
- Produced various financial metrics including physical value at risk, financial risk ratios, and absolute financial scores
- Performed UAT testing by writing unit tests and validation tests for a database of 25,000 records
- Developed a predictive model using sklearn to estimate companies future valuations under NGFS temperature scenarios

Project | Emissions Analysis

- Independently built a tool in Jupyter Notebook to calculate carbon emissions of private equity portfolios, including QA checks, a MinIO Cloud Database, and config driven development
- Developed an attribution tool in Jupyter Notebook to ascribe changes in a portfolio's carbon footprint specific factors: investment decisions or company-level decisions

Water Resource Data Analyst | The Water Center at Penn | Philadelphia, PA

Oct. 2020 – May 2022

- Analyzed 1,500 bacteria samples for the Schuylkill and Delaware River using Python and Excel, resulting in the discovery of new locations for public access to the river and a new theory on combined sewer overflows
- Wrote various statistical tests and produced over 50 graphs in python to identify trends between bacteria and rainfall

Sustainability Data Analyst | Penn Sustainability | Philadelphia, PA

June 2021 – May 2022

- Analyzed 600,000+ purchased records within Excel to quantify Scope 3 supply chain carbon emissions with the goal of reducing Penn's carbon emissions to net zero by 2042, resulting in the discovery of 5 major hot spots to target and reduce
- Wrote a 23-page report (background, methodology, and results) on Penn's Scope 3 emissions following the GHG Protocol

Financial Consultant | First Derivatives | New York City, NY

June 2018 – Feb. 2020

- Facilitated the Liquidity Stress Testing project for ScotiaBank as the lead UAT tester for BAU monthly code releases
- Inspected a total of 21,000 client records by collecting verification documentation for new compliance standards for BNY Mellon

EDUCATION

University of Pennsylvania | Master of Environmental Studies | Philadelphia, PA

May 2022

Concentration in Resilience and Adaptation, GPA 4.0/4.0

Project | NYC Inundation using Remote Sensing

- Used Sentinel-1 SAR imagery and Google Earth Engine to identify flood inundation across New York City following extreme rainfall events using JavaScript

Stony Brook University | BSc (Hons) Applied Mathematics and Statistics | Stony Brook, NY

May 2018

Concentration in Operations Research, Minor in Music, GPA 3.7/4.0, Magna Cum Laude

CERTIFICATIONS

IBM Data Science Professional Certificate

March 2025

IBM Machine Learning Certificate

August 2025

DeepLearning.AI Mathematics for Machine Learning and Data Science Certificate

August 2025

SKILLS

- Python, Excel, Git, Google VM and Google Cloud